VISHWAS · SIH26079 · TEAM HELLFIRE LABS

Paras

Metrics and evaluation integrity

Final prep — script, judge Q&A, and every number traced

BCA | Slide 4 — first half | about 65 seconds of speaking, down from 1m40

YOUR JOB

You have the slide that decides whether the judges believe the rest. Feasibility, the numbers, and — unusually — the three defects we found in our own work. Deliver those defects without apology: they are the strongest credibility signal on the deck. Viability and the risks have moved to Arnav, who follows you on the same slide; every question about metrics, statistics or the defects still comes back to you.

1 RUNNING ORDER

Mahika leads. Your rows are highlighted.

#	SPEAKER	WHAT HAPPENS	TIME
1	Mahika	Slide 1 — opens. Team, problem statement, the hook.	~40s
2	Arnav	Slide 2 — what VISHWAS is, and why the forecast underneath decays.	~70s
3	Devansh	Slide 2 — the measurement, and the result.	~60s
4	Ankit	Slide 3 — opens it: what an ensemble is, and why we score against rain gauges.	~35s
5	Devansh	Slide 3 — the pipeline, the stack, features, label, model, the map.	~75s
6	Mahika	Bridge into slide 4.	~10s
7	**Paras**	**Slide 4 — feasibility, the numbers, the three defects.**	**~65s**
8	Arnav	Slide 4 — viability, the four risks, the precision choice.	~40s
9	Mahika	Slide 5 — the three interviews.	~45s
10	Dhruv	Slide 5 — who it serves, MoES's numbers, the benefits. Then slide 6 — sources.	~75s
11	Devansh	Slide 6 — our own measurement.	~20s
12	Mahika	Closes. The tagline, then into questions.	~20s

2 THE STORY — EVERYONE MUST BE ABLE TO SAY THIS

If a judge stops you in the corridor. Thirty seconds. Learn the shape, not the words.

Every weather forecast India issues arrives with no honesty label.

Forecasters judge confidence today by ensemble spread — run the model 11 times from slightly different starting conditions, and if the runs disagree, distrust the forecast. We measured that method against India's own rain gauges, across all four seasons and three years.

Compared crudely it looks strong: ensemble spread appears to explain 43.9% of the variance in forecast error. Compared fairly — within the same rainfall conditions — it explains under 1%.

So we built the replacement. On that same fair comparison it scores 0.565 and VISHWAS scores 0.728, on a test year the model had never seen.

Most teams pitch a hypothesis. We pitch a measurement.

3 WHAT CHANGED IN YOUR SCRIPT

Your wording is your own. Only these points moved.

- Q&A OWNERSHIP — Devansh takes What is your accuracy. Your PR-AUC answer stays, but it is for the metric argument rather than the number, and there is a note on it now.
- Q&A RECHECKED — your significance answer still called one test year a floor and said the slide agreed. The slide now says the opposite: it was thin, so we retested. Devansh and Arnav both deliver that retest; you were the one walking it back. Fixed.
- You also have a new question: does the edge survive to Day 10. It does, at every lead day, but it narrows — do not say it is as wide at Day 10 as at Day 1.
- SPLIT — your script is about a third shorter. Viability, the four risk cards and the precision design choice have moved to Arnav, who picks up straight after you. You keep feasibility, the numbers, the 189,758 tie and the three defects, because those are your job and nobody else can answer for them. You end on “Ab Arnav aapko viability aur risks batayenge” and stop.
- Because the risks moved, their questions moved too. One test year, land-only domain, why GEFS and not NEPS, could this run at NCMRWF — Arnav takes those now. Every question about metrics, statistics, leakage or the defects is still yours, and you take it back even if it lands while Arnav is speaking.
- “VISHWAS scored 0.728 compared to the baseline 0.565” now says which comparison it is. Unqualified, a judge may assume it is the pooled number — and the entire pitch turns on the difference between the crude and the fair comparison.
- The margin and the 95% interval added. You had the point scores but not the statistics behind them.
- The 189,758 tie added. It is the caption printed on your own figure, it is arguably the sharpest fact in the deck, and nobody was saying it.
- Everything else is exactly as you wrote it — but tightened. The opening questions, the “let’s start with feasibility” line, the explanation of why we report mistakes, the cry-wolf mechanism and the closing summary all came out. None of it is lost: the first two were scaffolding Mahika’s bridge already covers, and the rest live in your Q&A. Your original was running about 2 minutes 20. After the tightening and the split with Arnav it is about 65 seconds — and every number on it is still yours.

4 YOUR SCRIPT

Grey boxes are stage directions — do not say them. Lines tagged TRIM are safe to cut if you are running long.

SLIDE 4 — FIRST HALF

▸ *Mahika bridges into you: "...and the three defects we found in our own work, and fixed."*

Thank you, Mahika.

Feasibility pehle. VISHWAS ko humne 2018 par train kiya, aur test kiya 2019 ke saare 52 Wednesdays par — ek aisa saal jo model ne kabhi dekha hi nahi tha.

Fair comparison par — matlab same rainfall conditions ke andar — baseline ka score 0.565 hai, aur VISHWAS ka 0.728. Margin +0.13 se +0.16, 95% interval poora zero se upar, aur aathon rainfall categories mein aage.

▸ *PAUSE*

Operational terms mein: same number of alerts par VISHWAS 64% busts pakadta hai, current method 55%.

Aur ek baat. Test set ke lagbhag ek-tihai hisse par — 189,758 forecasts par — ensemble spread exactly 0.00 millimetre hai. Matlab current method unhe rank hi nahi kar sakta. VISHWAS us tie ke andar bhi rank karta hai.

▸ *BIG PAUSE — look at the judges. This is the strongest thing on your slide.*

Humne apne hi kaam mein teen defects nikaale aur fix kiye — do data leaks, aur ek baseline tie. Hum unhe openly report kar rahe hain.

Ab Arnav aapko viability aur risks batayenge.

▸ *Hand to Arnav — he takes the rest of Slide 4. You are done. Do not add a summary line after this.*

5 EVERY NUMBER YOU SAY, AND WHERE IT COMES FROM

If a judge challenges a figure, give the source — not a defence.

CLAIM	SOURCE / WHAT IT REALLY MEANS
0.565 vs 0.728, +0.13 to +0.16	Stratified ROC-AUC. 95% CI +0.108 to +0.193, cluster bootstrap over forecast dates.
all eight rainfall bins	Ahead by +0.079 to +0.252. EIGHT bins, not deciles.
64% vs 55%	POD 0.641 vs 0.546 at a matched 10% alert budget. Operating point chosen on validation, never on test.
three defects	Two leakage bugs plus the baseline tie — detail in your Q&A below.
189,758 (the caption under the figure)	Rows where ensemble spread is exactly 0.00 mm — a third of the test set, one tie it cannot rank at all.

6 QUESTIONS THE JUDGES WILL ASK YOU

Yours to answer. Read them out loud once — an answer you have said aloud comes back under pressure; one you have only read does not.

Q1 Why PR-AUC and not accuracy?

The base rate is about 6%. A model that always says ‘no bust’ gets 94% accuracy and is worthless. Accuracy on an imbalanced problem is a number that tells you nothing.

We report ROC-AUC within rainfall regime, PR-AUC lift over the base rate — 7.60 times for us against 6.29 for spread — and operational scores at a matched alert budget.

⚠ Devansh takes 'what is your accuracy' — let him. This is yours only if they press on WHY PR-AUC, which is the metric argument rather than the number.

Q2 Explain the matched alert budget.

We let both systems flag exactly the same fraction of forecasts — the riskiest 10% — and ask which catches more real busts. That removes the trivial way to win, which is simply to alert more often.

At 10%: VISHWAS gets POD 0.641, FAR 0.586, CSI 0.336, ETS 0.300, against ensemble spread's 0.546, 0.647, 0.273 and 0.235. Delta-POD is +0.093 with an interval of +0.071 to +0.110.

And the operating point was chosen on the validation slice, never on test.

Q3 What were the three defects?

One. The climatology behind our anomaly features was originally computed over training and test data together, so every test row's features encoded the test period's own average. Caught, fixed, and proven fixed — the old code had a signature, every group's anomalies summing to exactly zero.

Two. A second leak of identical shape in the bust threshold, pre-empted before it could be written.

Three, and this is the one I would not have found without checking. Our first operational comparison had the baseline flagging 38.78% of cases against a 10% alert budget — which silently voided the matched-budget premise. The cause is that ensemble spread is exactly 0.00 millimetres across 91% of dry cases, so a percentile cut flags everything at once. Fixed with rank-based selection and random tie-breaking.

⚠ Deliver this without apology. It is the reason to believe every other number on the deck.

Q4 Why bootstrap by date rather than by row?

Two gridpoints 50 kilometres apart on the same day are very nearly the same observation. Treating every row in the test set as independent would produce a confidence interval far too narrow — it would look impressive and be wrong.

So we resample whole forecast dates — 52 of them — because the independent unit is the forecast, not the gridpoint.

Q5 Why weight false alarms above missed events?

Two reasons. The theme is Smart Automation, and an automated assistant that cries wolf gets switched off — and then it helps nobody. That effect is documented; LeClerc and Joslyn, Risk Analysis, 2015.

And we heard it directly in our own field research: an NDRF constable told us he had mobilised for rain that never came.

Q6 How do I know there is no leakage now?

Four things. The label threshold is computed on the training window only, then frozen and applied to test. The anomaly climatology is fitted on training only. No feature is derived from the truth — that separation is enforced.

And the test base rate is deliberately not exactly 5%. If it were, that would be the tell that the threshold had seen test data.

Q7 Eight bins — why not ten?

The code asks for deciles of forecast rainfall, but about 32% of test rows have a forecast of exactly 0.00 millimetres, so three quantile edges coincide and get dropped. The realised count is eight.

Bin zero absorbs the whole zero block — 208,065 rows at a 0.15% bust rate, against the wettest bin's 40.92%. I say eight bins rather than deciles because eight is what actually happened.

Q8 How many wet-point definitions did you try?

Five, and the model is ahead under all of them. The honest margin range is +0.132 to +0.163, taken from the variants that condition only on information available at forecast time.

Our table also has truth-filtered variants, and we exclude those from the range deliberately. Requiring the truth to be wet deletes every ‘forecast said dry, it poured’ bust, and requiring the forecast to be wet deletes every ‘forecast said rain, nothing came’ bust. Both extreme classes vanish and only middling busts remain, so the margin shrinks for reasons that have nothing to do with model quality.

⚠ Never quote a margin range containing +0.035. That is the truth-filtered one.

Q9 What is the negative result you found?

An ensemble-size test showed the baseline improves slightly when you go from 5 to 11 members — which makes our own claim marginally weaker, and it is written down anyway.

Going from 11 to 31 cannot be measured from this archive, and we say that too rather than assuming it away.

Q10 What are POD, FAR, CSI and ETS?

Probability of detection: of the real busts, how many we caught. False alarm ratio: of the alerts we raised, how many were wrong. Critical success index and equitable threat score combine the two, with ETS correcting for hits you would have got by chance.

They are the WMO standard definitions — Jolliffe and Stephenson — so they are the metrics an operational centre already uses to judge a forecast product.

Q11 Does the edge survive to Day 10?

Yes, at every lead day, with the whole confidence interval above zero at all ten.

It does narrow gently across the range — about +0.171 at Day 1 easing to +0.149 at Day 10 on the two-year test. Positive throughout is the claim I will stand behind.

⚠ DO NOT say the margin is as wide at Day 10 as at Day 1. That rested on the old 52-date curve and the 104-date curve does not support it.

Q12 Is the improvement statistically significant?

The 95% interval on the margin is +0.108 to +0.193, entirely above zero, and the interval on delta-POD is +0.071 to +0.110, also entirely above zero.

And that interval is within one test year, so we went and tested a second one: retrained on 2015 to 2017, rescored on 104 forecast dates across two monsoons, margin +0.149 with the whole interval above zero. The slide says so — it no longer calls one test year a floor.

7 IF IT IS NOT YOURS, HAND IT OVER

Naming the right teammate reads as a team that knows its own work. Guessing reads as one that does not.

IF THEY ASK ABOUT…	SAY THE NAME
Viability, the risks, why GEFS and not NEPS, one test year, land-only	Arnav
What the model is, how it works internally	Devansh
Where the data came from, the pipeline, the fetch	Devansh
What an ensemble is, why gauges rather than reanalysis	Ankit
A real-world bust example	Arnav
The interviews	Mahika
Sources, impact, roadmap	Dhruv

SITUATION	WHAT TO SAY
You don't know the answer	“I don't have that number to hand and I'd rather not guess — can I come back to you on it?” That is a strength for this team, not a weakness: when we could not trace one of our own figures, we removed it rather than present it.
It's a teammate's area	“That's <name>'s part — <name>, do you want to take that?” Say the name. Don't answer for them.
A judge challenges a number	Give the source, not a defence. Every figure on the deck traces to a government source, a peer-reviewed paper, or our own logs.
A judge says 'you're just predicting heavy rain'	Anyone can answer this one: we removed the rainfall forecast from the model entirely and the score went UP, 0.728 to 0.731.

8 NEVER SAY THIS

Every one of these was a real error caught during the build, or a rule set deliberately.

- × Never say '43.9% of forecast error'. It is 43.9% of the VARIANCE in forecast error. This is the single easiest thing to get wrong under pressure.
- × Never quote the margin as '+0.163'. Say '+0.13 to +0.16'.
- × Never say 'deciles' or 'the wettest tenth' about the model result. It is EIGHT bins. (Deciles are correct only for the separate Phase 1 correlation analysis.)
- × Never quote any accuracy figure. Base rate is about 6%, so accuracy is meaningless — and quoting it signals a team that does not understand imbalance.
- × Never say 'AI-adjusted forecast' or imply we correct the forecast. We DETECT busts. We do not correct anything.
- × Never say the pooled ROC-AUC pair (0.909 vs 0.885) — not on a slide, not in a figure, not aloud, not even if a judge asks directly. Answer with the pooled MARGIN, +0.023, which is published on slide 4. Quoting the pair as performance would commit, in our own work, exactly the error our pitch accuses the incumbent of.
- × Never quote +0.663 as 'the baseline we beat', and never quote a margin range containing +0.035 (that one came from a filter conditioning on the outcome).
- × Never call the regions 'IMD official regions'. They are approximate lat/lon boxes.
- × Never present the confidence map's date as evidence of general performance. It is ONE date, illustrating the form of the product.
- × Never convert the three field interviews into percentages or 'N% of users said'. They are three exploratory conversations.
- × Never quote 98%→46%, ₹13,331 cr, '50% faster', '~30% improvement', or a 0.34–0.37 false-alarm ratio. All were in earlier drafts; none could be traced; all were removed. Do not say any of these numbers out loud even while explaining that you removed them.
- × Never say SHAP and permutation importance 'agree on the drivers'. They pick the same top three in a different order.

9 NUMBER CHEAT SHEET

Same in all six packs. If two of you quote a figure differently in the same room, that is the damage.

WHAT	NUMBER	WATCH OUT
Spread explains, pooled	43.9% of the VARIANCE in forecast error	not '43.9% of forecast error'
Spread explains, fair comparison	0.17%–0.90% → say 'under 1%'	never quote +0.041 alone
VISHWAS score (stratified ROC-AUC)	0.728	—
Ensemble spread score	0.565	—
Margin	+0.13 to +0.16 · 95% CI +0.108 to +0.193	never '+0.163' alone
Matched 10% alert budget	VISHWAS catches 64% of busts vs 55%	—
POD / FAR / CSI / ETS at 10%	0.641 / 0.586 / 0.336 / 0.300 vs 0.546 / 0.647 / 0.273 / 0.235	—
Features	23, all from GEFS, none from the truth	not 24
Ensemble members	11 (control + 10 perturbed)	—

WHAT	NUMBER	WATCH OUT
Domain	1,252 gauge points of a 5,041-point analysis box (24.8%)	the 5,041 box is land AND ocean; only the 1,252 is land
Data fetched	50.3 GB · 104 forecast dates (2018-2019)	this study only; both studies together are 124.5 GB over 256 dates
Labelled rows	1,302,080 in this study · 651,040 in the test set	don't conflate the two; across both studies it is 3,205,120
Five-year retest	margin +0.149, 95% CI +0.116 to +0.203, 104 forecast dates	quote the MARGIN only, never an AUC beside 0.728
Live cycle scored	GEFS 20260907 and 20260908, ~100 s, ~210 MB, CPU	the live run is DONE — never call it future work
High-risk share	7.78% gauge-verified · 14.06% unverified part of the shipped map	ALWAYS name the domain; never blend them
Train / test	trained on 2018 · tested on all 52 Wednesdays of 2019	—
Base rate	about 6%	so never quote accuracy
Bust definition	|forecast − observed| above the 95th percentile, per region and lead time, threshold set on training data only	—
Rainfall bins	EIGHT bins · 40.92% bust rate in the wettest, 0.15% in the driest	never 'deciles' or 'wettest tenth'
Ablations	full 0.728 · no month 0.722 · no rainfall forecast 0.731 · no 850 shear 0.726 · spread-only 0.562	—
Underlying GEFS, heavy rain (≥64.5 mm)	detects 23% of events at Day 1, 0.2% at Day 10	this is the forecast, NOT our model
IMD warning accuracy (MoES)	85, 73, 67, 63, 58% for Day 1–5	Lok Sabha Q.2953, 17 Dec 2025
The tie	ensemble spread is exactly 0.00 mm on 189,758 test rows — a third of the set	—
Calibration	Brier 0.04217 vs 0.04749 · resolution 1.4× the incumbent's	—

ONE LAST THING

The strongest thing about this project is not the score. It is that we measured the method used today before proposing a replacement, and that we removed our own figures when we could not source them. If you land one impression, land that one.